MIL-P-13041E
31 January 1969
SUPERSEDING
MIL-P-0013041D(MU)
7 February 1968, and
MIL-P-13041C
31 March 1967

MILITARY SPECIFICATION

PROJECTILE, 4.2 INCH,

SMOKE, M328A1;

WP FILLING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the white phosphorus (WP) filling and assembly of the M328A1 projectile, without fuzing components.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-215 - Chemical Agent, White Phosphorus, (WP).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 1315

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DRAWINGS

US ARMY MUNITIONS COMMAND

8797838 - Projectile, 4.2 Inch, WP, Smoke, M328A1, Loading Assembly.
8848553 - Pallet for 4.2 Inch Mortar Projectile Bodies.
8861520 - Marking Diagram for Pallets for Projectile 75MM to 120MM.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing 8797838 or on the subsidiary drawings shall conform to the drawings and specifications listed thereon or to the specific characteristics set forth on the drawings.

3.2 Filling. The projectile, clean and free from foreign matter, shall be filled with molten WP conforming to MIL-C-215, at a temperature of $140^{\circ} \pm 10^{\circ}$ F (see 6.2 and 6.3). When water is in contact with the WP during the filling operation, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. After test for leakage, the WP shall be solidified with the projectile in a nose-up position.

3.3 Assembly. Closing, coating, marking and assembly of the projectile shall be in accordance with Drawing 8797838.

3.4 Filling height. There shall be a 3 to 5 percent void, by total volume, existing above the surface of the water over the WP in the filled and closed projectile. The depth of water remaining in the filled and closed projectile shall not exceed 1/4 inch. The projectile shall be radiographed in a vertical, nose-up position while the temperature of the WP is $130^{\circ} \pm 10^{\circ}$ F when tested as specified in 4.2.4.2 (see 6.3).

3.5 Cleaning after closing. The exterior surface of the filled and closed projectile, and the interior of the burster casing, shall be thoroughly cleaned to remove any grease, oil, WP or other contamination (see 6.4).

3.6 Leakage. The closed and cleaned projectile shall not leak when maintained at 207° to 212° F for a minimum period of 15 minutes. Alternate method: the projectile shall not leak when maintained at $160^{\circ} \pm 5^{\circ}$ F for a period of 8 hours. The projectile shall be tested as specified in 4.2.4.1.

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3.7 Workmanship. The projectile shall be free from internal and external contamination (foreign matter) and damage such as dents, cracks, or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the filled and closed projectiles, of one interfix number, produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and offered for delivery at one time. However, no more than one weight zone nor more than one metal parts interfix number shall be represented in any one lot.

4.2.2 Sampling for examination and test. Sampling shall be conducted in accordance with MIL-STD-105. For radiographic inspection, general inspection level I shall be used for the first three consecutive acceptable lots; thereafter, special inspection level S-4 shall be used.

4.2.3 Inspection procedure.

4.2.3.1 For examination and test. Sample projectiles shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For critical defects. Each projectile shall be inspected for the critical characteristics listed in the classification of defects.

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4.2.3.3 Classification of defects

(a) Projectile, 4.2 inch, WP, smoke M328A1, loading assembly (Drawing 8797838).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.2.4.1
2	WP contamination (3.5)	
Major:	AQL 1.0 percent defective	
101	Filling excessive	4.2.4.2
102	Excess water	4.2.4.2
103	Filling insufficient	4.2.4.2
104	Burster casing not sealed	
105	Bourrelet diameter (maximum) after coating	
106	Marking incorrect	
107	Weight zone incorrect	
108	Damage (to threads internal and/or external and/or body) (3.7)	
Minor:	AQL 2.5 percent defective	
201	Marking missing	
*202	Internal contamination (3.7)	
203	Coating missing, incorrect or inadequate	
204	Contamination after filling (3.7)	
	AQL 10.0 percent defective	
205	Marking illegible	

*To be checked prior to filling.

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4.2.3.3 Classification of defects (continued).

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Pallets
102	Protective sleeves missing
103	Strapping loose or missing
104	Marking illegible or incorrect

4.2.4 Tests.

4.2.4.1 Leakage. The projectiles shall be placed in an oven in the nose-up position for the 207° to 212° F method (for the alternate method, in the nose-down position at 160° ± 5° F) and maintained at the specified temperature for the required period of time and examined for leakage. The munition temperature shall be measured by thermocouples inserted in the burster wells of at least two munitions located in the center of each oven bay batch. The emission of smoke constitutes leakage.

4.2.4.2 Filling height. The filled and closed projectile shall be radiographed while the WP is molten as specified with the projectile in a nose-up position and inspected to assure proper filling height.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. Eighty-three filled and closed projectiles per pallet shall be packed in accordance with Drawing F8848553.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Drawing C8861520.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 Caution. Because of the hazardous nature of WP, filling installations are advised to take all possible precautions to protect personnel from injury while filling projectiles.

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6.3 Filling aid. Projectiles filled with approximately 8.14 pounds of WP have been found to comply with 3.4. This information is furnished as an aid to procurement and not as a criterion for filling.

6.4 Decontamination. The filled and closed projectiles may be decontaminated by passing through a hot water and detergent spray washer or by flushing with hot water while vigorously scrubbing with steel wool. Additional cleaning of the burster well may be effected with a steam lance and a power driven wire brush.

Custodians:

Army - MU
Air Force - 70

Preparing activity:

Army - MU(EA)

Review activity:

Air Force - 70

Project No. 1315-0661

User interest:

Navy - MC

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